Examiner only

7. *(a)* Describe the main features of the spectrum of a star and state where in the star they arise. [2]

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(b) The table gives some information about two stars.

Star	Luminosity / W	Distance from Earth / m
Sirius	9.7×10^{27}	8.1×10^{16}
Vega	1.5×10^{28}	2.4×10^{17}

(i) Determine the ratio:

$$\frac{\text{Intensity of radiation reaching Earth from Sirius}}{\text{Intensity of radiation reaching Earth from Vega}}$$ [3]

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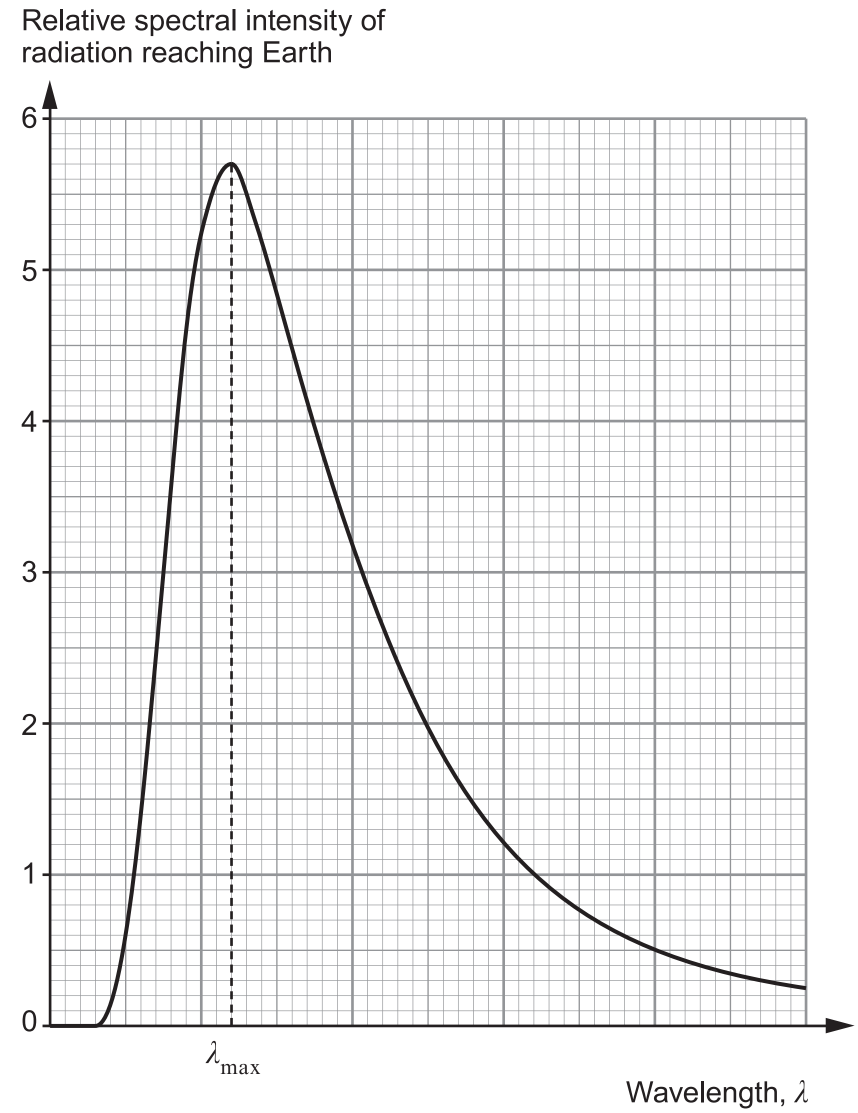
Examiner only

(ii) The two stars have similar surface temperatures. Below is a black body radiation curve for Sirius. Using your answer to *(b)*(i), sketch the expected black body curve for Vega on the same axes. [2]



(iii) Determine λ_{max}, given that the surface area of Sirius is $1.8 \times 10^{19}\,m^2$. [4]

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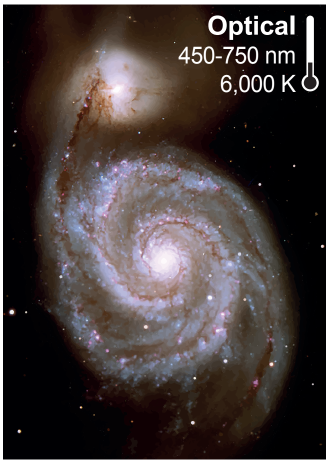
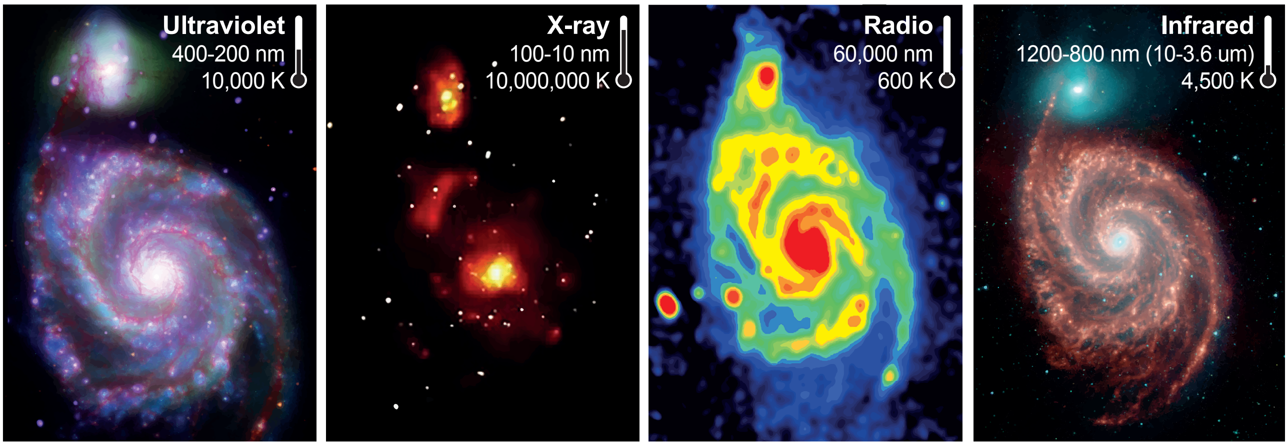
(c) The image below is of the whirlpool galaxy, M51 (or NGC 5194). This is one of the first galaxies to be photographed by astronomers.



Subsequent images of the **same galaxy** are shown below.



Describe how these developments in observational astronomy have advanced the study of the whirlpool galaxy. [3]

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END OF PAPER